



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

Spring 2025 Exam

ST443

Machine Learning and Data Mining

Suitable for all ST candidates

Instructions to candidates

This paper contains four questions. Answer **ALL FOUR**.

Question 1: 20 marks

Question 2: 30 marks

Question 3: 25 marks

Question 4: 25 marks

The marks in brackets reflect marks for each part of a question.

Candidates must not take the question paper away with them after the examination. Please leave it on your desk at the end of the examination.

Time allowed	Reading Time:	<i>None</i>
	Writing Time:	<i>2 hours</i>

You are supplied with:	<i>No additional materials</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are allowed in this exam</i>
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1. Are the following statements true or false? Provide a brief explanation.

(a) The following classifier is a linear classifier:

$$\hat{\psi}(x) = 2 \times \mathbb{1}(G(x^\top \beta) \geq 0) - 1 \text{ for } x \in \mathbb{R}^p$$

where $\beta \in \mathbb{R}^p$ is a parameter, $G : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing and continuous function, $\mathbb{1}(\cdot)$ is the indicator function, and the two class labels are -1 and 1 . [2 marks]

- (b) k -NN classifiers are fitted by using stochastic gradient descent algorithm to minimise the loss function. [2 marks]
- (c) The AIC is evaluated using a validation set of data points. [2 marks]
- (d) In K -fold cross-validation, each data point belongs to exactly one test fold; therefore, the test folds are independent. Consequently, the test error estimates of the separate folds are mutually independent. [2 marks]
- (e) There can be an infinite number of parameter vectors that minimise the sum of the squared prediction error and the Ridge penalty function, given the regularisation hyperparameter $\lambda > 0$. [2 marks]
- (f) The forward stepwise selection searches through at most cp models, where c is some constant and p is the dimensionality of the feature vectors. [2 marks]
- (g) Ridge regression is used for variable selection. [2 marks]
- (h) A decision tree generated with binary splitting can have the same number of internal and terminal nodes. [2 marks]
- (i) Random forests can provide an improvement over bagged trees because they tend to produce more correlated trees. [2 marks]
- (j) The decision boundary produced by a support vector machine (SVM) is determined solely by the support vectors. [2 marks]

2. This problem concerns the Bayes risk with respect to the pin-ball loss function $\ell : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$, defined as follows:

$$\ell(y, t) = \begin{cases} (1 - \tau)(t - y), & \text{if } y < t \\ \tau(y - t), & \text{if } y \geq t \end{cases}$$

where y is the true value, t is the predicted value, and τ is a parameter with value in $(0, 1)$.

Assume that for every x , $\mathbb{P}[Y \leq y \mid X = x]$ is a continuous function in y .

- (a) Show what the Bayes risk minimiser $\psi_\tau(x)$ is. [15 marks]
- (b) Discuss the interpretation of the Bayes risk minimiser in relation to the conditional distribution of Y given $X = x$, and the type of regression performed using the pin-ball loss function. [5 marks]
- (c) Assume that the conditional distribution of Y given $X = x$ is a uniform distribution on $[a_x, b_x]$, where $a_x < b_x$. Show that the Bayes risk is

$$R^* = \frac{1}{2} \tau (1 - \tau) \mathbb{E}[b_X - a_X].$$

[10 marks]

3. This problem concerns gradient boosting for regression using the squared-error loss function for the input data points $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R} \times \mathbb{R}$ such that $x_1 < x_2 < \dots < x_n$. Consider the gradient boosting algorithm at iteration m . Recall that $f_{m-1}(\cdot)$ denotes the ensemble of boosted trees computed through iterations $1, \dots, m-1$ of the algorithm. At iteration m , a decision tree is selected. Assume that the algorithm chooses a depth-1 decision tree as a weak learner, using the variance gain selection criterion.
- When we refer to a tree that splits at k , we mean that the tree partitions the points into two regions: $R_1 = \{x \in \mathbb{R} : x \leq x_k\}$ and $R_2 = \{x \in \mathbb{R} : x > x_k\}$.
- For a tree that splits at k , for $1 \leq k < n$, what is the output of the tree for any given x ? [5 marks]
 - Assume that $\sum_{i=1}^n y_i = \sum_{i=1}^n f_{m-1}(x_i)$. Show that it is more beneficial to use a tree that splits at $k = 1$ than at $k = n - 1$ if $|y_1 - f_{m-1}(x_1)| > |y_n - f_{m-1}(x_n)|$. [10 marks]
 - Assume that $\sum_{i=1}^n y_i = \sum_{i=1}^n f_{m-1}(x_i)$, and, for some constant $c \neq 0$, $\sum_{i=1}^k y_i - \sum_{i=1}^k f_{m-1}(x_i) = c$, for every $k = 1, \dots, n-1$. What are the optimal splits? [10 marks]
4. This problem concerns k -means clustering. You are given a dataset containing the annual incomes (in thousands of GBP) and years of experience for a group of employees. The dataset is provided in Table 1.

Table 1: Dataset for Problem 4.

Employee ID	Income (in 1000 GBP)	Years of Experience
1	40	2
2	80	8
3	45	3
4	75	6
5	100	10
6	90	9
7	50	4
8	65	7

You are tasked with using k -means clustering to classify employees into two groups based on their income and years of experience.

Assume that the initial cluster centroids correspond to employees 1 and 5 for centroids 1 and 2, respectively.

- Briefly describe the steps involved in applying the k -means algorithm to this dataset. Assume that we are clustering the data into 2 clusters ($k=2$).
- Using the initial centroids, calculate which cluster each data point would be assigned to after the first assignment step. Show your work. [5 marks]
- After the assignment step, how would you update the centroids of the clusters? Calculate the new centroids. [5 marks]
- What can you infer from the clusters about the employees' income and years of experience? [5 marks]
- Describe how the steps involved in computing the clustering will be changed if we used the k -means++ algorithm instead. [5 marks]